

PANUWAT SANGKETKIT

Robotics & Control Systems Engineer

Yokohama, JAPAN

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PROFESSIONAL SUMMARY

Robotics & Control Systems Engineer with 7+ years of experience in industrial automation and real-time control systems, with additional experience in embedded software development and automotive ECU validation. Specialized in motion control, sensor integration (IMU, distance sensors), and communication systems (CAN, TCP/IP, PLC-PC). Proven track record of deploying reliable systems in field environments.

WORK EXPERIENCE

System Engineer

Nov 2021 – Present

TOTO ELECTRIC INDUSTRY CO., LTD.

Yokohama, Japan

- Architected and implemented real-time control software for an automated concrete spraying robot using VB.NET, focusing on high-precision motion and system reliability.
- Developed advanced motion control algorithms, including vertical/horizontal interpolation and surface-following logic, which enhanced spraying stability and operational efficiency.
- Integrated multi-sensor fusion (IMU and distance sensors) to enable dynamic path correction, allowing the robot to adapt to complex surfaces in real-time.
- Spearheaded the PLC HMI migration project from Keyence to Schneider/Pro-face, improving the user interface for field operators and streamlining system maintainability.
- Built TCP/IP communication between PC and PLC (Keyence), improving system responsiveness.

Software Engineer

Oct 2019 – Jul 2021

EXARS CORPORATION

Aichi, Japan

- Built Simulink validation tools using MATLAB (m-files).
- Built database communication systems using MySQL and C#.
- Developed embedded and control software for automotive systems using C, C#, and Python.
- Developed ECU testing tools using CAPL and C programming.
- Created internal automation tools using VBA to improve workflow efficiency.

R&D Engineer

Feb 2018 – Sep 2019

Aimagin Co., Ltd.

Samut Prakan, Thailand

- Developed embedded systems using ESP32 (C programming).
- Designed and implemented handheld measurement devices (Heavy metal detection systems).
- Designed circuit schematics and PCBs for embedded hardware.
- Provided technical support for Simulink-based system development.

SKILLS

Programming:

C, C#, VB.NET, Python, VBA

Robotics & Control Systems:

Robot Control, Motion Control, Path Planning,
Interpolation, Kinematics, Kawasaki AS,
Closed-loop control, PID tuning

Real-Time & System Engineering:

Sensor Integration (IMU, Distance Sensors), Real-time Processing, Control Algorithms

Embedded & Automotive Systems:

CAN Communication, ECU Testing

Industrial Automation:

PLC System Integration, HMI Design,
PC-PLC Communication

Tools & Technologies:

Git, MySQL, MATLAB/Simulink, WebRTC,
Socket.io, Visual Studio, VS Code, Keyence PLC,
Pro-face (Schneider)

Languages:

English (Professional Working Proficiency),
Japanese (JLPT N3, N2 Candidate for July 2026),
Thai (Native)

CERTIFICATIONS

Industrial Robot Teaching (Teach Pendant) & Safety Training

Mar 2022

Kawasaki Robotics, Japan

PROJECTS

Wearable ECG Monitoring Device

2015 – 2017

King Mongkut's University of Technology Thonburi (KMUTT), Thailand

- Developed a low-complexity dual-threshold QRS detection algorithm for real-time wearable ECG devices
- Evaluated algorithm robustness under noise and motion conditions using synthetic and real ECG datasets.
- Achieved reliable detection performance at SNR ≥ 12 dB, suitable for low-power embedded systems.
- Designed end-to-end system including BLE-based data transmission to Android devices.
- Published research in IEEE conference: <https://ieeexplore.ieee.org/document/8096215>

Assistive Laundry-Carrying Robot for Elderly

2013

King Mongkut's University of Technology Thonburi (KMUTT), Thailand

- Developed a mobile robot to assist elderly users in carrying laundry.
- Implemented target detection using ultrasonic sensors for distance and direction tracking.
- Integrated digital compass for orientation control.

ADDITIONAL EXPERIENCE

Research Intern

Jul 2013 - Dec 2013

National Electronics and Computer Technology (NECTEC)

Pathum Thani, Thailand

Rehabilitation & Assistive Technology Lab

- Designed PCB circuits for assistive robotic systems
- Assembled and wired robotic hardware components for prototype development

EDUCATION

Master of Engineering (M.Eng.) in Control Systems & Instrumentation Engineering

2014 – 2018

Bachelor of Engineering (B.Eng.) in Control Systems & Instrumentation Engineering

2011 – 2014

King Mongkut's University of Technology Thonburi (KMUTT), Thailand